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Automatic Billing With Object Detection and Counting

Bharat Singh, Mh. Salim, Nidhi Kushwaha

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ABSTRACT

In the realm of small businesses, such as restaurants, artisanal shops, fresh produce vendors, and customized product retailers, the absence of barcode scanning systems often leads to significant billing challenges. These establishments, characterized by a wide array of unique and handmade items, face inefficiencies in their billing processes, resulting in slower service, potential errors in transaction records, and an increased risk of financial discrepancies. The need for a sophisticated yet accessible billing solution is paramount to ensure accuracy, security, and efficiency in transactions, catering specifically to the diverse inventory these businesses possess. To address these challenges and revolutionize the billing experience for both small business owners and their customers, we introduce the AI Powered Smart Shopper Automatic Billing with Product Classification and Measurement. This innovative system harnesses advanced machine learning techniques to develop a model adept at recognizing and classifying a broad spectrum of products typically found in small business settings. By integrating state-of-the-art image recognition and quantity measurement algorithms, the AI-powered Smart Shopper system is uniquely capable of accurately identifying and categorizing items, thereby facilitating an automated billing process. The implementation of this system marks a significant departure from traditional manual cash handling and cashier-dependent transactions. Furthermore, this approach alleviates the need for direct physical contact with money, aligning with contemporary preferences for hygiene and safety. By leveraging machine learning and image recognition technologies, the AI Powered Smart Shopper system promises to deliver a seamless, accurate, and secure billing experience. It not only addresses the inherent problem of billing inefficiencies in small businesses but also offers a tailored solution that enhances operational efficiency, improves transaction accuracy, and provides peace of mind for business owners. Through the adoption of this system, small businesses can look forward to a future where billing processes are no longer a bottleneck but a catalyst for improved customer satisfaction and business growth.

INTRODUCTION

Automating the billing process not only enhances security by minimizing human error and potential for theft but also substantially streamlines operations, enabling businesses to serve their customers more efficiently

1.1 Motivation

The motivation behind this project is that the current retail billing systems are slow and error-prone due to manual scanning and human intervention, leading to inefficiencies at checkout points.

With the retail landscape demanding faster and more reliable service, there is a critical need for automated solutions. Automatic billing systems using object detection can drastically cut checkout times and reduce errors by recognizing and counting items like apples, bananas, and samosas automatically.

Adopting such technologies not only streamlines operations, reducing labor costs and improving inventory management, but also enhances customer satisfaction. It paves the way for broader application of machine learning and computer vision in retail.

Implementing an automatic billing system addresses major inefficiencies in retail, offering a significant improvement in both operational efficiency and customer experience, and advancing the integration of intelligent systems in commerce.

1.2 Objectives of the Project

The objectives of this project are:

1. To develop customized software solutions for small businesses with the aim of automating their operations and improving their profitability.
2. The primary focus is on creating tailored software systems that eliminate the need for barcodes and enable 24-hour operations, thereby reducing costs and increasing revenue potential.
3. Develop custom software solutions for specific small businesses, designed to automate various aspects of their operations, such as billing, inventory management, and customer interactions.
4. Design software that eliminates the reliance on barcode systems, thereby reducing the cost of implementing and maintaining barcodes. Explore alternative methods, such as AI-powered image recognition, to streamline product identification and pricing.
5. Develop software features that enable businesses to operate 24 hours a day. Assess the feasibility and benefits of extended hours, including increased customer accessibility and potential revenue gains.
6. Develop software features that enable businesses to operate 24 hours a day. Assess the feasibility and benefits of extended hours, including increased customer accessibility and potential revenue gains.

LITRETURE SURVEY

S.No.	Paper Title and Author name	Proposed Methods	Advantages	Disadvantage
26.	Intelligent Billing system using Object Detection. Author: --Neeraj Chidella, N Kalyan Reddy, N Sai Dheeraj Reddy, Maddi Mohan 2022	The proposed system aims to significantly reduce the time required for billing, especially for fruits and vegetables by automating the process of detection and billing.	The paper uses a fine-tuned CNN for single fruit detection, achieving an accuracy of 98 percent. CNN is employed for classifying and identifying individual fruits in images.	Disadvantage of the paper is that it focuses on object detection and billing for a specific set of fruits, vegetables, and some grocery items. While the results seem promising for this limited set of classes, the system may not be easily adaptable to a broader range of objects without significant retraining and dataset expansion.
1.	Smart shopping system with an RFID Interface for Human. Author-R.R. Vallabhuni, S.Lakshmanachari, G. Avanthi and V.Vija. 2021	Using an RFID tag for generating a bill.	It is fast compared to the barcode scanner.	Adding RFID to product is an additional time-consuming task

1.	Survey on object detection techniques Author: - Shrikant A. Shinde1, Akshay Mali2, Aniket Thorat3, Saurabh Khot 2022	The paper utilizes YOLO for object recognition, allowing the cart to identify and classify products placed in it.	Using YOLO for product recognition and sensors, the cart automatically identifies products placed in it by customers.	The reliance on multiple sensors and hardware components in the proposed smart cart system can lead to increased cost, technical challenges, potential reliability issues, scalability concerns, and a learning curve for customers
30.	Object Detection for Inventory Stock Counting Using YOLOv5 Author: - Isaiah Francis E. Babila, Shawn Anthonie E. Villazor, Jennifer C. Dela Cruz 2022	The paper presents a solution for automating inventory management in the cell phone business using object detection algorithms, offering improved accuracy, time savings, and reduced reliance on manual counting processes	The YOLO algorithm enables real-time detection of objects, making it suitable for tracking inventory in dynamic environments. the system aims to achieve higher accuracy in counting and identifying specific products	One potential disadvantage of the study is the reliance on specific lighting conditions, as highlighted in the "Lighting Condition" section. The study determined that the best lighting source for accurate object detection was a 9W light source. This means that the system's performance may be sensitive to lighting conditions, and variations in lighting could potentially impact the accuracy of object detection
5.	Grab, Pay and Eat: Semantic Food Detection for Smart Restaurants Author-Eduardo Aguilar, Beatriz Remeseiro, Marc Bolanos, and Petia Radeva 2020	This paper addresses the problem of automatic food tray analysis in canteens and restaurants. This method integrates food localization, recognition, and segmentation into a single framework based on convolutional neural networks (CNNs).	This paper help in monitoring food consumption and automating the billing process. This is particularly advantageous in highdemand scenarios, reducing the cashier's workload.	One potential disadvantage of the proposed system is that it may still require some manual intervention or correction in cases where the system's predictions are not accurate
48.	Improved YOLOv5: Efficient Object Detection Using Drone Images under Various Conditions Hyun-Ki Jung * and Gi-Sang Choi 2022	The paper aims to improve object detection performance in difficult environmental and weather conditions using drones.	By modifying the Conv layer and replacing the SiLU activation function with ELU helps achieve better performance, particularly in adverse situations, such as during heavy rain or low visibility.	Modifying the Conv layer and activation functions, as mentioned in the paper, could potentially increase the computational demands of the model, making it less feasible for deployment on resource constrained drones or edge devices

13 3	A Comparative Assessment of Deep Neural Network Models for Detecting Obstacles in the Real Time Aerial Railway Track Images. R.S. Rampriyaa, R. Suganyab, Sabari Nathanc, and P. Shunmuga Perumal. 2022	The study focuses on detecting obstacles on railway tracks using aerial images captured by UAVs. To facilitate this, a new dataset called the Rail Obstacle Detection Dataset (RODD) is created, consisting of annotated images with obstacle information	The proposed method allows for real-time monitoring of railway tracks using UAVs, enabling early obstacle detection. The study also evaluates a range of deep neural network models to identify the most effective approach for obstacle detection.	The literature survey mentions the impact of different lighting conditions on model performance. Real-world scenarios often involve varying lighting conditions and adverse weather
17 40 5	Object detection using YOLO: challenges, architectural successors, datasets and applications Tausif Diwan 1 & G. Anirudh 2 & Jitendra V. Tembhurne 2022	This paper focus on Examination of architectural advancements within single-stage detectors, encompassing network design, feature extraction, and other components impacting their performance	The paper reviews single stage object detectors, with a focus on YOLO and its successors	The paper mentions YOLO versions up to YOLOv4, but it does not include information on more recent developments, like YOLOv5. Staying up-to-date with the latest research is essential in a rapidly evolving field like computer vision
19 22 54 25 34	You Only Look Once: Unified, Real-Time Object Detection. Joseph Redmon, Santosh Divvala, Ross Girshick, Ali Farhadi 2016	The key idea behind YOLO is to treat object detection as a regression problem, directly predicting the bounding boxes and associated class probabilities from image pixels.	YOLO is exceptionally fast, capable of processing images in real-time at a high frame rate (45 fps for the base model and over 150 fps for a faster version). This makes it suitable for applications like autonomous driving and real-time scene analysis.	YOLO imposes spatial constraints on bounding box predictions, as each grid cell can only predict two boxes and one class. This limitation may hinder its ability to detect multiple nearby objects, especially small objects in groups.
55 5	Object Detection through Modified YOLO Neural Network anvir Ahmad , 1 Yinglong Ma , 1 Muhammad Yahya,2 Belal Ahmad, 3 Shah Nazir , 4 and Amin ul Haq 2020	The proposed method introduces a new loss function that is more flexible and optimized. It uses proportionality instead of a fixed difference to address issues with neighbouring objects and small object detection.	More flexible and optimized loss function, which is better suited for detecting both large and small objects	The confusion matrix analysis reveals that certain object categories are easily confused due to their visual similarity. model might not perform well in scenarios where visually similar objects need to be accurately distinguished.

Table1: Literature survey

DESIGN AND METHODOLOGY

3.1 Architecture Diagram

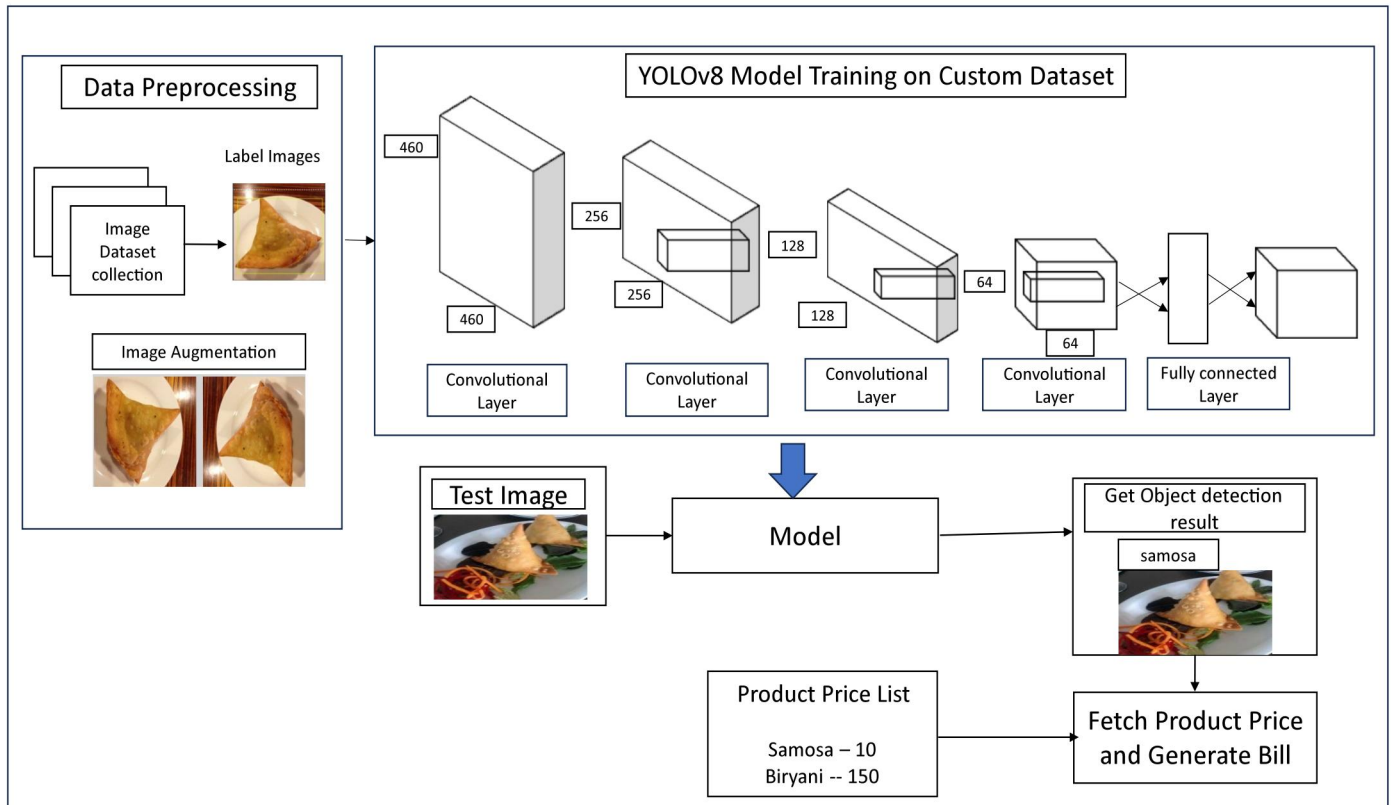


Fig.1: Architecture diagram

Following data preparation, the focus shifts to training a **YOLO (You Only Look Once) model**, specifically tailored to recognize samosas from the custom dataset. This step is pivotal as the YOLO model, renowned for its efficiency in **object detection**, is trained with images enriched with labels and bounding box coordinates to grasp the spatial dynamics between objects and their classifications.

The culmination of this project is the bill generation phase. After training, the YOLO model is employed to analyse test images, accurately identifying and counting the samosas within. It then retrieves pricing details for each detected object from a predefined database. The final bill is computed by aggregating the costs of all identified items, thus automating the billing process based on object detection.

3.2 Module Description

- 1) Data Acquisition and Annotation:** We began by sourcing a diverse set of samosa images from Kaggle, amassing a total of 1023 images. To facilitate supervised learning, each image was meticulously annotated with bounding boxes outlining the samosas within them. This annotation process was streamlined using Roboflow, ensuring accuracy and consistency across the dataset.

3.3 Model Training

- 1) YOLOv8 :** YOLOv8 is the latest iteration in the series of **YOLO (You Only Look Once)** models, known for their speed and accuracy in detecting objects in images and videos in real-time. YOLOv8, like its predecessors, is designed to be extremely fast, making it suitable for applications requiring real-time processing.

Architecture and Algorithms:

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Unified Architecture: YOLOv8 continues the tradition of a unified architecture that predicts bounding boxes and class probabilities in a single forward pass of the network, making it incredibly fast.

Backbone: It uses a more advanced and efficient backbone for feature extraction, which may include improvements on CSP (Cross Stage Partial) networks, or other efficient architectures that enhance feature learning while maintaining high processing speed.

Head: The prediction head of YOLOv8 has likely been optimized further for better handling of scale variation and aspect ratios, improving detection of small or challenging objects.

43

Algorithms: It incorporates state-of-the-art techniques in object detection, including improved loss functions for better bounding box regression, class imbalance handling, and potentially the use of transformers or attention mechanisms to enhance the model's understanding of complex scenes.

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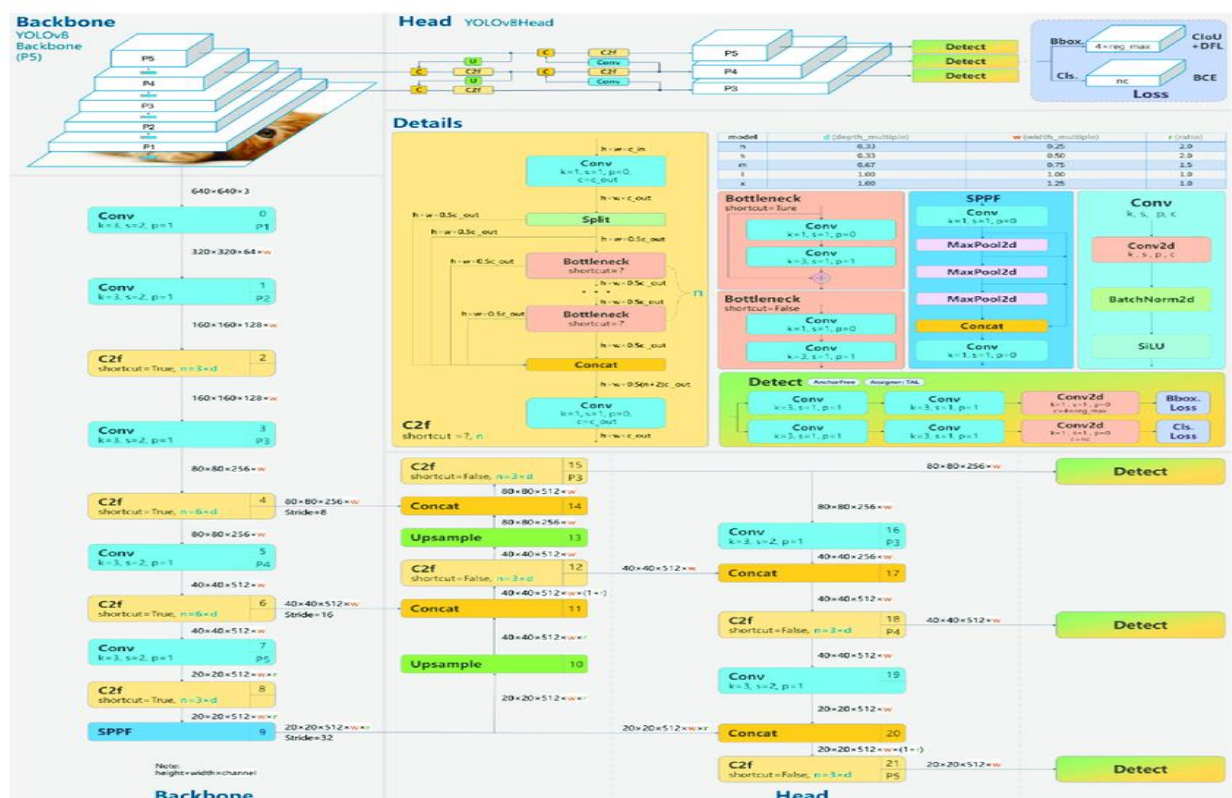


Fig.7: YOLOv8 Architecture

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2) YOLOv5: YOLOv5, despite the versioning, is an evolution of the YOLO architecture that focuses on being lightweight, fast, and easy to deploy. It's widely used in various applications for its balance between speed and accuracy.

Architecture and Algorithms:

Simplicity and Speed: YOLOv5 offers a range of models (from YOLOv5s to YOLOv5x) with varying sizes to suit different computational budgets and accuracy needs.

Backbone: Utilizes a CSP-based backbone for efficient feature extraction, focusing on reducing computational complexity while maintaining high performance.

Head: The detection head uses anchor boxes, with improvements in predicting object bounds and class probabilities. It employs techniques like CIoU loss for better bounding box regression.

Augmentations and Techniques: YOLOv5 incorporates extensive data augmentation techniques and training strategies like mosaic augmentation, auto-learning bounding box anchors, and model pruning for deployment.

3) Faster R-CNN: Faster R-CNN (Region-based Convolutional Neural Network) is a highly influential model in the field of object detection, known for its accuracy. It introduces the concept of Region Proposal Networks (RPN) for generating object proposals.

Architecture and Algorithms:

Two-Stage Detection: Faster R-CNN operates in two stages. The first stage, RPN, proposes candidate object bounding boxes. The second stage uses these proposals to classify object types and refine their bounding boxes.

Backbone: It can use various backbone architectures like VGG, ResNet, or more recent networks for feature extraction. The choice of backbone affects the model's accuracy and speed.

Region Proposal Network: RPN is a fully convolutional network that efficiently predicts object bounds and object scores at each position. It uses anchor boxes to accommodate different object sizes and aspect ratios.

RoI Pooling: The Region of Interest (RoI) Pooling layer extracts a fixed-size feature vector from each proposal for classification and bounding box regression, ensuring that features are aligned with the proposed regions.

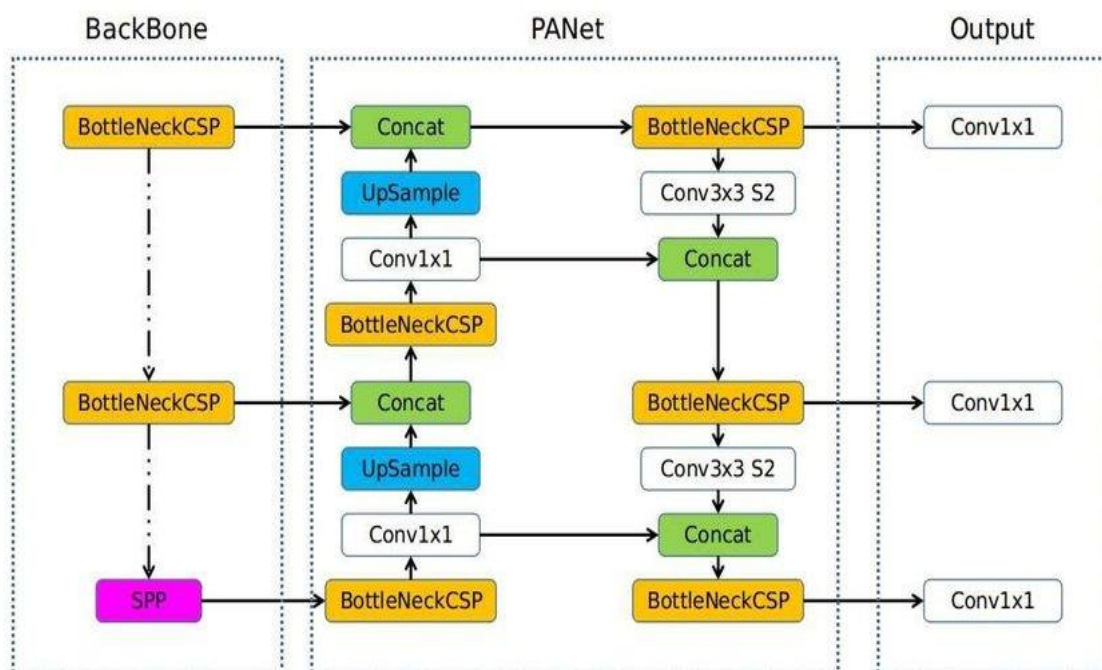


Fig.8: YOLOv5 Architecture

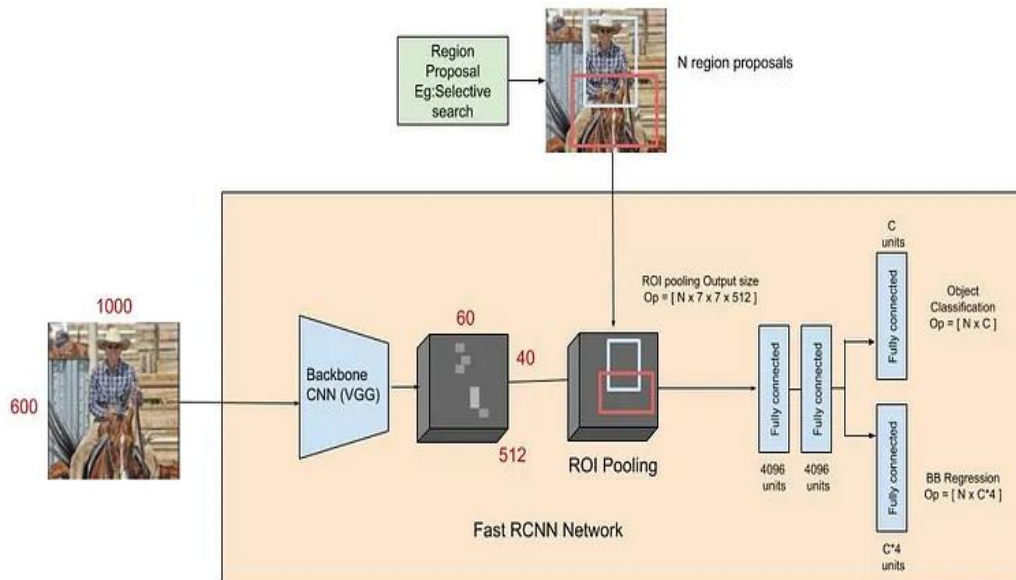


Fig.9: Faster RCNN Architecture

3.4 Object Detection and Counting

- Object Detection:** Initially, images are passed through a custom YOLOv8 (You Only Look Once) trained model. YOLO is a popular algorithm for object detection that can accurately identify and localize objects within images. This step focuses on identifying various objects present in the image, such as products on store shelves.
- Object Tracking:** After object detection, object tracking is performed using techniques like DeepSORT (Deep Simple Online and Realtime Tracking). Object tracking helps in keeping track of individual objects over time, especially in scenarios where objects might move or change position within the scene.
- Counting Objects:** While object tracking provides information about the movement and location of objects, accurate counting of objects is crucial for automatic billing. To achieve this, a weight measurement technique is employed. Each object has a fixed weight associated with it. By measuring the total weight of all detected objects and dividing it by the weight of a single object, the system accurately determines the count of objects present in the scene.

SIMULATION RESULT AND DISCUSSION

4.1 Models Performance and Matrix:

Model Name	Precision	Recall	Accuracy
YOLOV8	86	82	91
YOLOV5	88	85	91
Faster RCNN	35	98	45

Table.2: Models Performance Matrix

YOLOv8: -

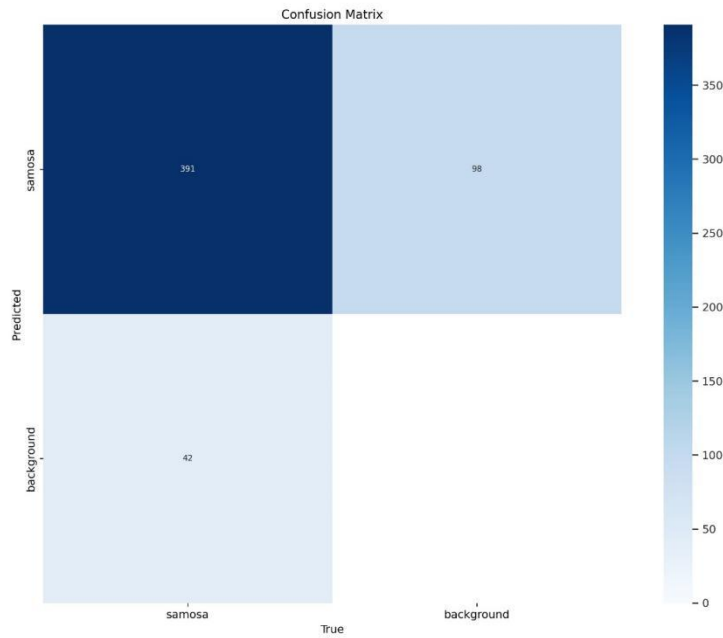


Fig.18: YOLOv8 Confusion Matrix

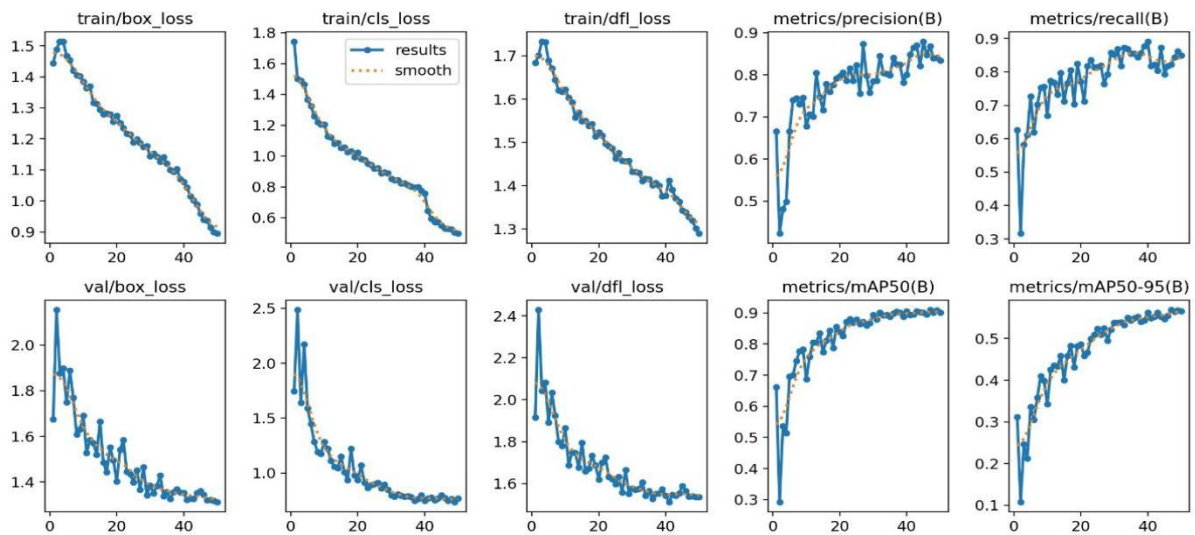


Fig.19: YOLOv8 training and validation loss

4.2 Performance Matrix of Various Classes:

Class Name	Precision	Recall	Accuracy
Banana	1	94	96
Samosa	88	85	90
Apple	71	73	78
Broccoli	57	51	52
Carrot	41	39	33

Table.3: Performance Matrix of Various Classes

Banana:

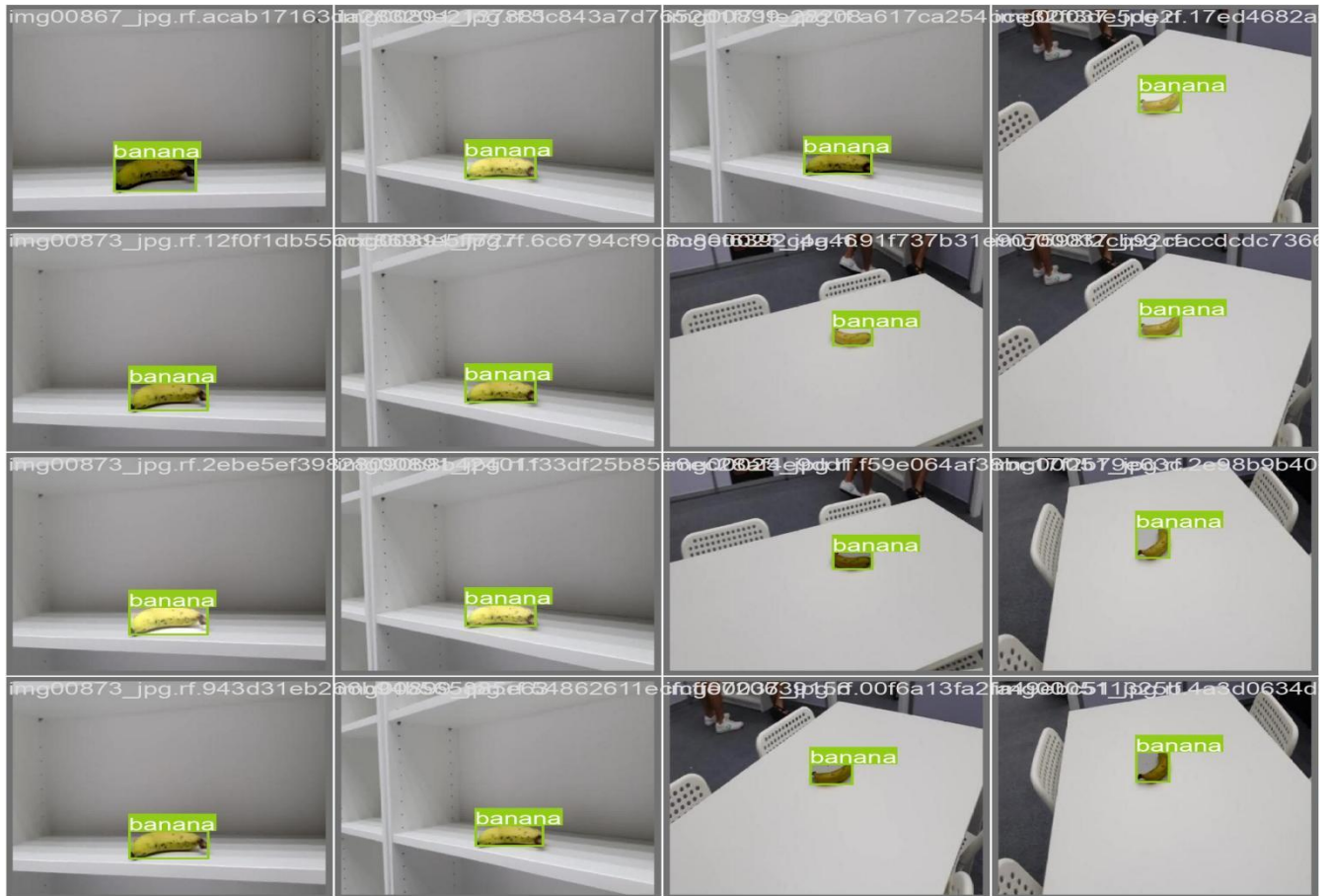
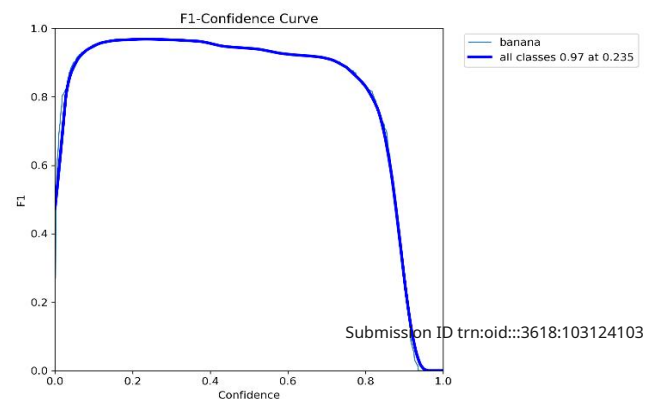
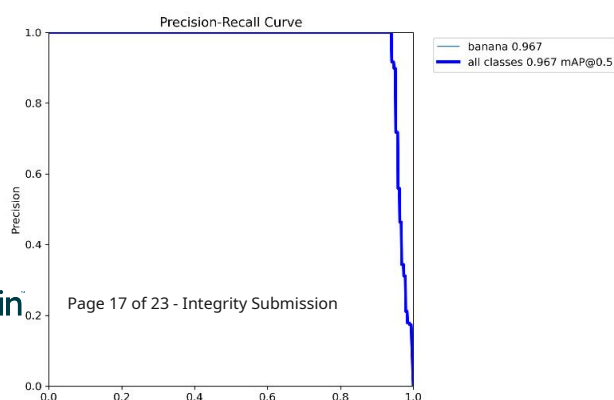
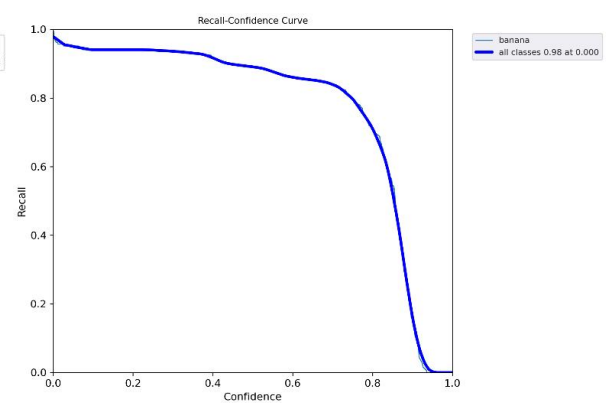
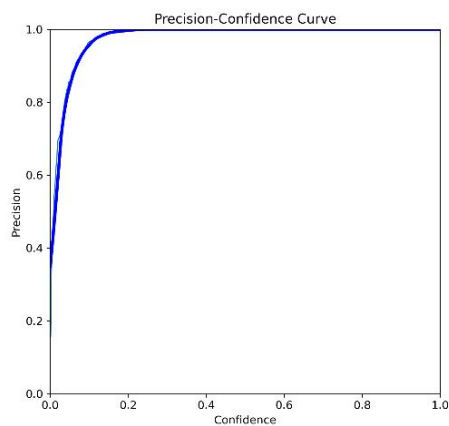


Fig.22: Banana Detection



Samosa:

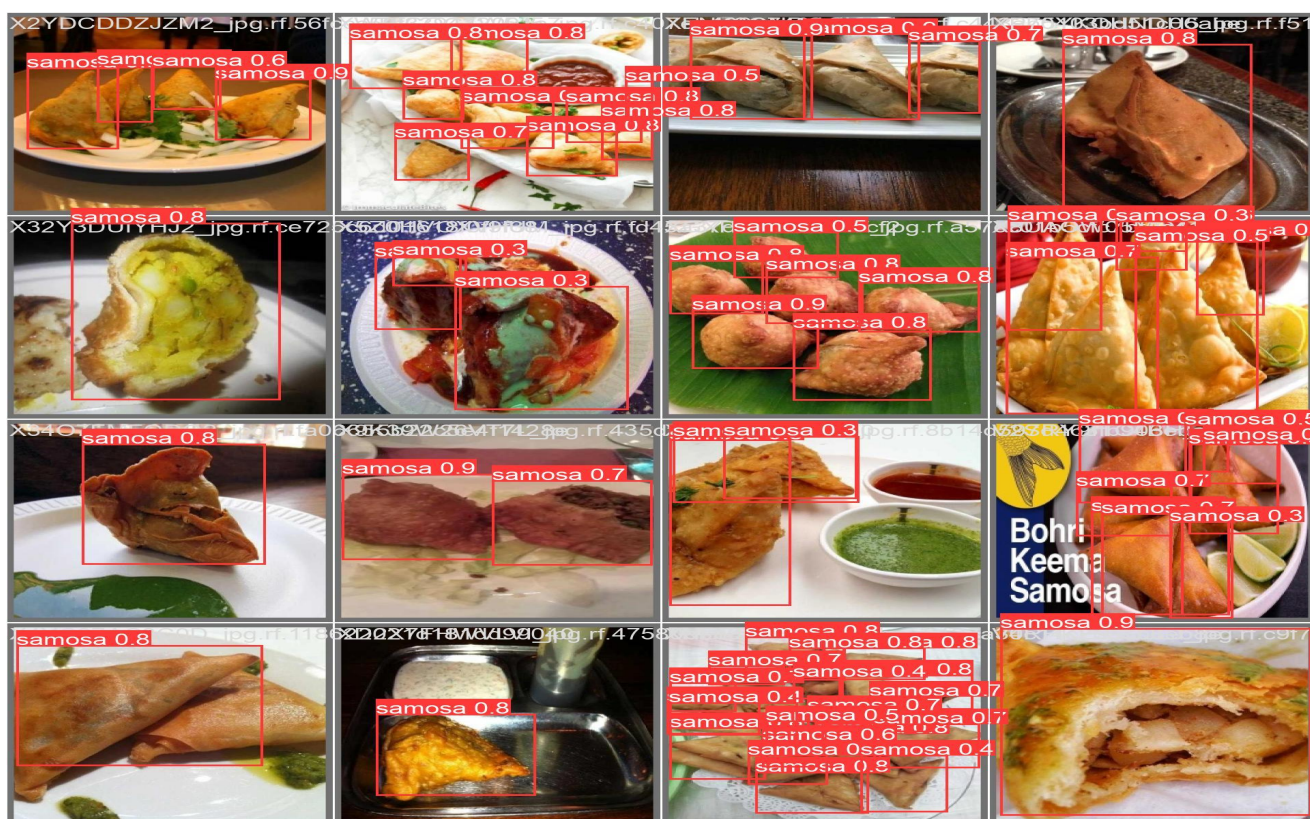
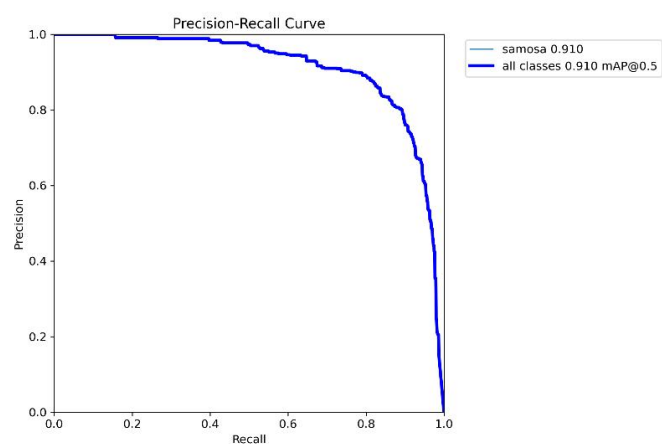
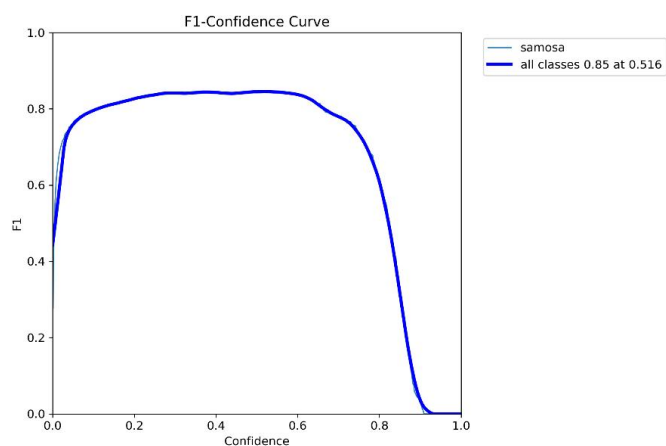


Fig.23: Samosa Detection



Apple:

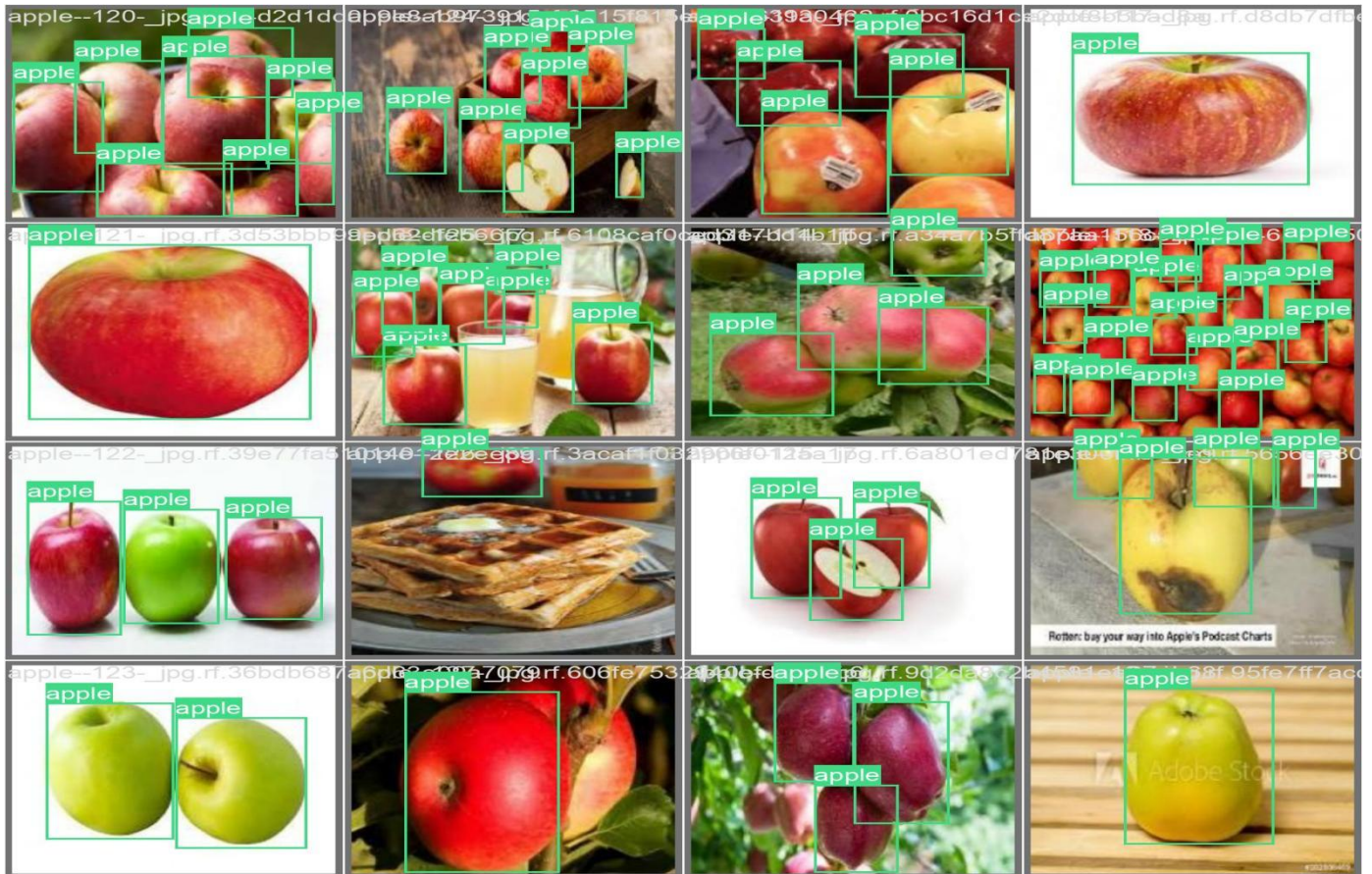
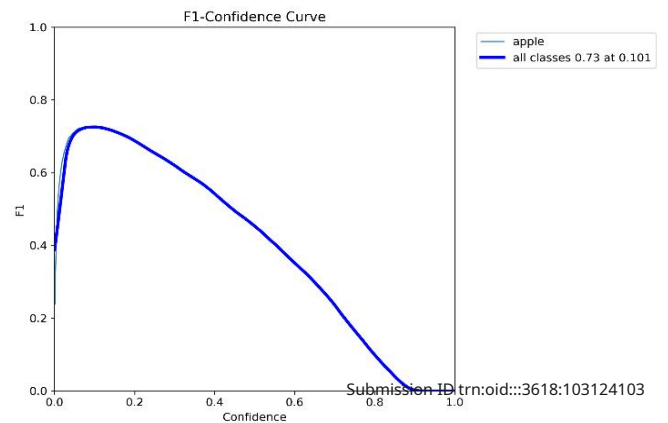
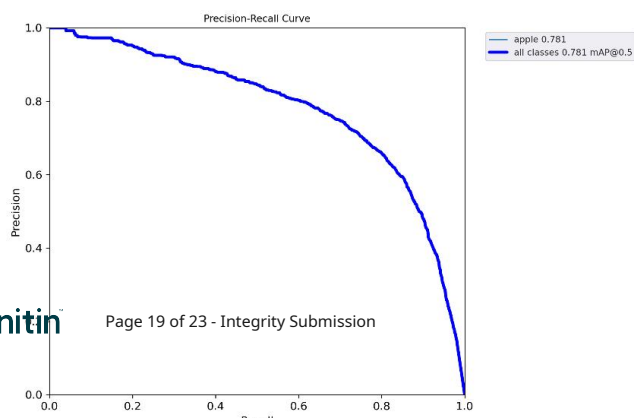
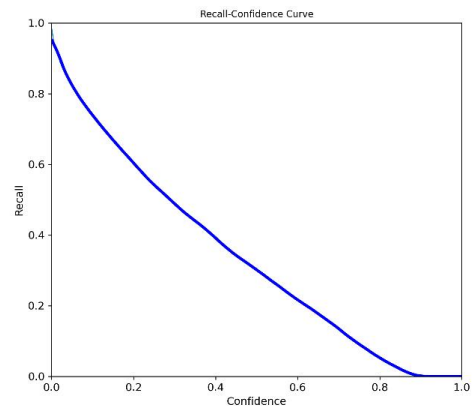
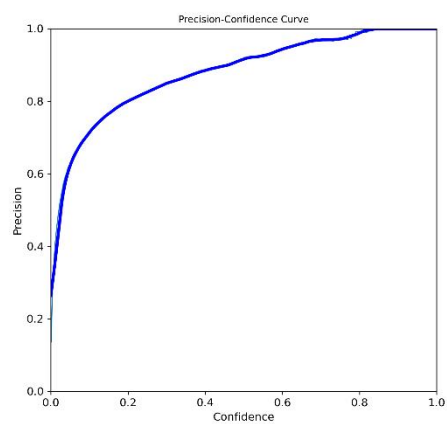


Fig.24: Apple Detection



Broccoli:

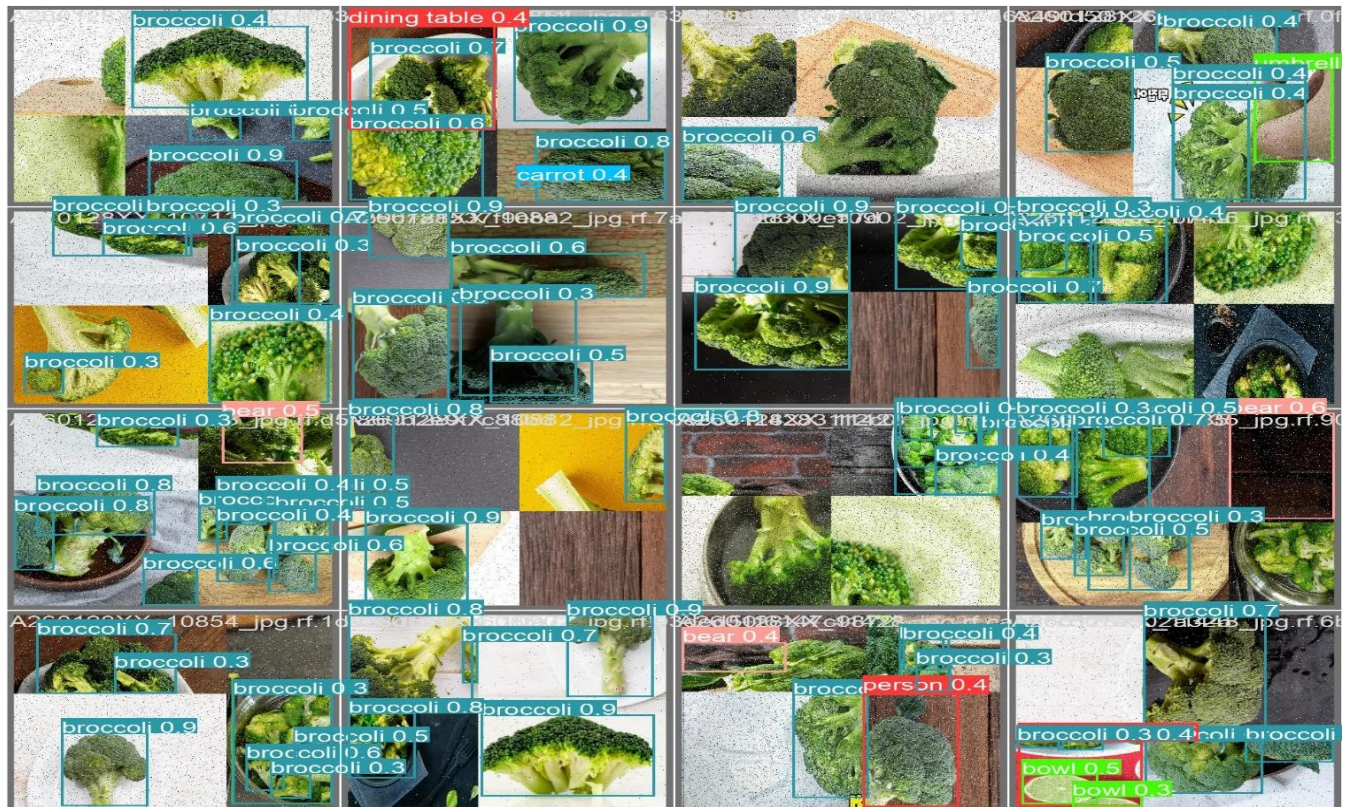
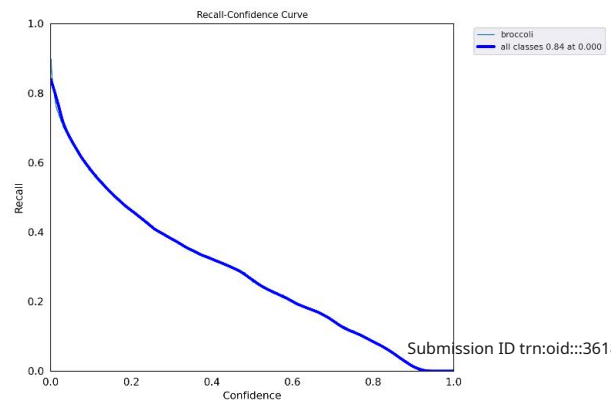
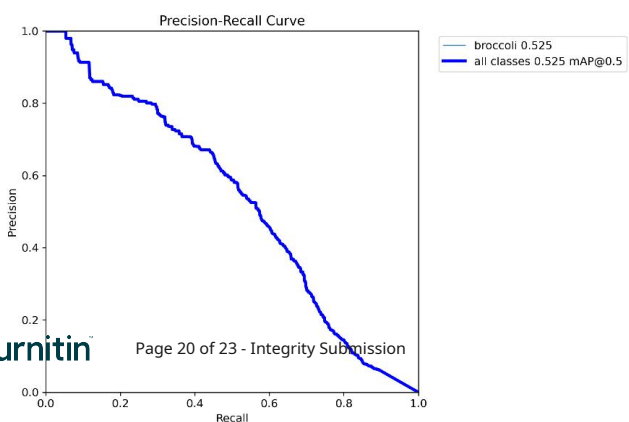
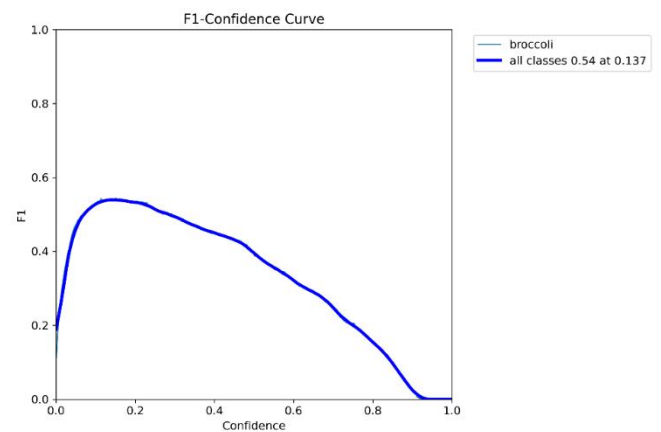
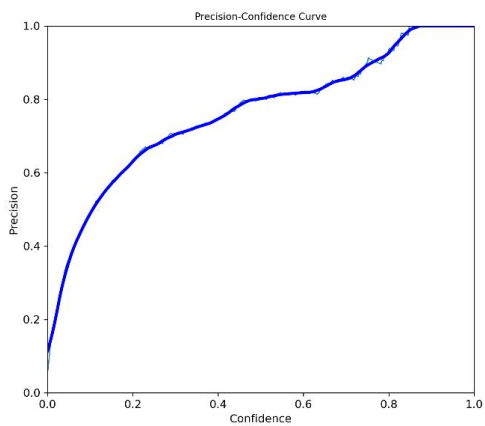


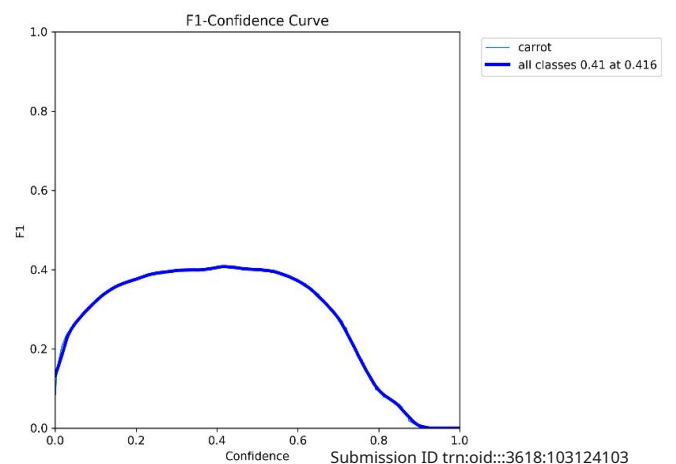
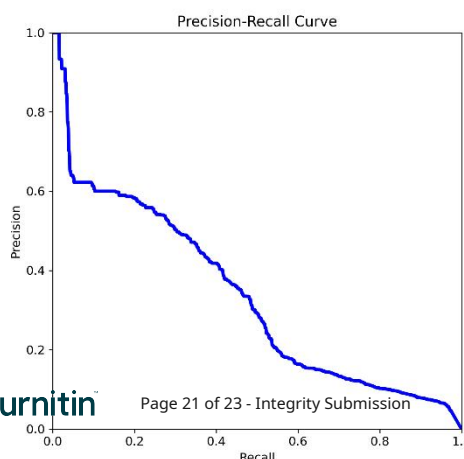
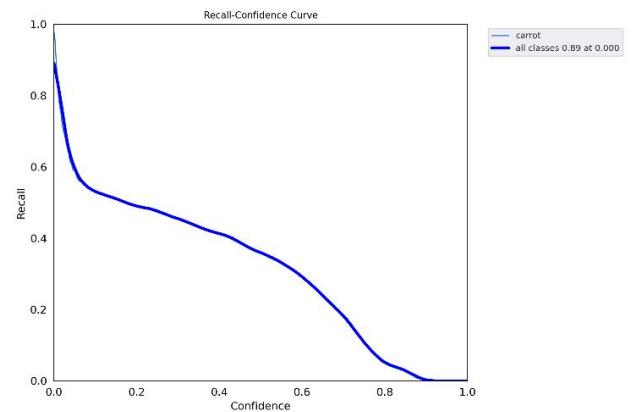
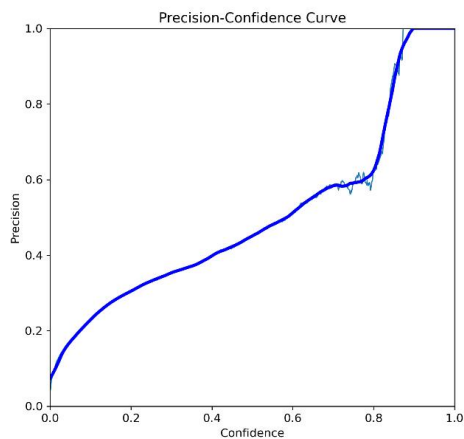
Fig.25: Broccoli Detection



Carrot:



Fig.26: Carrot Detection



CONCLUSION AND FUTURE SCOPE

- 44 Our research demonstrates the potential of AI-powered object detection for revolutionizing billing processes in small businesses. The proposed "Automatic Billing With Object Detection and Counting" system achieved promising results with a YOLOv8 model exhibiting an accuracy of 91%. This indicates the system's capability to streamline billing and enhance operational efficiency. While YOLOv8 outperformed YOLOv5 and Faster R-CNN in our experiments, future work will involve exploring alternative deep learning architectures and fine-tuning hyperparameters to potentially achieve even better accuracy. Additionally, we plan to significantly expand the model's object recognition capabilities to encompass a broader variety of products commonly found in diverse small retail settings. This will involve collecting and annotating a much larger dataset encompassing a wider range of product categories. Furthermore, we aim to integrate the system with existing point-of-sale (POS) systems for seamless integration into existing workflows. Finally, we will conduct user studies to evaluate the system's usability and gather feedback from both business owners and customers to ensure its practical adoption in real-world scenarios. By addressing these future directions, we envision the AI-powered smart shopper system becoming a widely adopted solution for enhancing billing efficiency and security in small businesses. Our future work focuses on enhancing the capabilities and real-world applicability of the "Automatic Billing With Object Detection and Counting" system. we will significantly expand the model's object recognition capabilities by collecting and annotating a much larger dataset encompassing a wider variety of products commonly found in diverse small retail settings. This will allow the system to handle a broader range of businesses and product types. Additionally, we aim to integrate the system seamlessly into existing workflows by enabling smooth interaction with point-of-sale (POS) systems used by many businesses.
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